

VIRTUAL PRIVATE CLOUD (VPC) CONFIGURATION

(Internet Gateway)

❖ What is an Internet Gateway?

- An **Internet Gateway (IGW)** is a **component of an AWS VPC (Virtual Private Cloud)** that enables **communication between instances in your VPC and the internet.**

➤ Key Functions of an Internet Gateway:

1. **Allows inbound traffic from the internet** to your public instances (if security rules allow).
2. **Allows outbound traffic from your instances** to the internet.
3. Performs **NAT (Network Address Translation)** for instances that have public IPs or Elastic IPs.
4. Scales horizontally—no bandwidth constraints from AWS

❖ Role of IGW in Internet Access

- The main job of an Internet Gateway is to enable EC2 instances in a VPC to communicate with the internet.

✓ ***This communication works in two directions:***

1. **Outbound Traffic:**

- Allows EC2 instances in public subnets to **send traffic to the internet** (e.g., access updates, connect to APIs). APIs, use cloud repositories, etc.

2. **Inbound Traffic:**

- Enables external users (e.g., browser clients) to **connect to public-facing EC2 instances** (e.g., web servers)

Note:-

1. *Users from the internet can access the instance (e.g., access a web server via HTTP, or a Linux server via SSH) – only if allowed by security settings and public IP.*
2. *But just having an IGW is not enough. Certain other components need to work together to enable full internet access.*

❖ **Public vs Private EC2 Instances**

- In AWS networking, EC2 instances can be classified as public or private, based on whether they can communicate with the internet directly.

▪ **Public EC2 Instances**

▪ ***These are instances that are:***

- Placed in a **subnet that has a route** to the Internet Gateway
- Assigned a **public IP address or Elastic IP**

▪ **Purpose:**

- Public instances are used when the instance needs to be accessed **directly from the internet.**

Examples:

- Web servers (e.g., hosting websites)
- Application frontends
- Load balancers
- Bastion hosts (for accessing private instances securely)

• **Private EC2 Instances**

• ***Private instances are those:***

- Placed in a subnet that does not have a route to the IGW
- Do not have a public IP
- ✓ *These instances cannot be accessed directly from the internet.*

Purpose:

They are used for internal systems that don't need internet exposure.

Examples:

- Databases (MySQL, PostgreSQL, MongoDB)

- Internal backend services
- Application logic servers
- Worker nodes

❖ **Combining Public and Private Instances (Best Practice Design)**

Layer/Component	Location	Purpose
Web Server (EC2)	Public Subnet	Receives traffic from the internet
Application Server	Private Subnet	Processes logic; not directly accessible
Database Server	Private Subnet	Stores data; highly
Internet Gateway	VPC	Public internet access for public subnet
NAT Gateway (optional)	Public Subnet	Enables internet access for private instances